

-

Intrusive and non-intrusive model reduction

Today

Today

- Organization
- Model reduction in a nutshell

Announcements

- Python/JAX recap in Exercise

Organization

- **Time and location:** Fridays, 08.15 - 10.00, MA A1 10
- **Exercise:** Fridays, 10.15 - 12.00, MA A1 10 (Attila Karsai)
- **Moodle:** <https://go.epfl.ch/MATH-537>

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 - Expect to discuss your homeworks/projects; write some code on the board.

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- Use of AI is **not** a requirement.
- You have to be able to explain every aspect of your homework/project/code in detail. You are responsible for every line of code and every answer you give.
- On each homework submission (not project), tell us some of the AI answers you *not* used and why, to demonstrate *how* you are using AI

Timetable of homeworks and projects

	posted	due	discussion
HW 1	18.09	16.10 at 8.00am	16.10 in exercise
HW 2	16.10	06.11 at 8.00am	06.11 in exercise
HW 3	06.11	20.11 at 8.00am	20.11 in exercise
Projects	20.11	11.12 at 8.00am	11.12 in class

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- Projects will be presented in groups in class on 11.12

Other

Prerequisites:

- Math-250: Advanced Numerical Analysis I
- Students should be comfortable with Python programming (preferably JAX)

Recommended:

- Math-351: Advanced Numerical Analysis II

Textbooks and reading material (not required)

- Jan S Hesthaven, Gianluigi Rozza, Benjamin Stamm, Certified Reduced Basis Methods for Parametrized Partial Differential Equations, Springer, 2016.
- Peter Benner, Serkan Gugercin, and Karen Willcox, A Survey of Projection-Based Model Reduction Methods for Parametric Dynamical Systems, SIAM, 2015.
- G. Rozza, D. B. P. Huynh, A. T. Patera, Reduced basis approximation and a posteriori error estimation for affinely parametrized elliptic coercive partial differential equations, Springer, 2007.

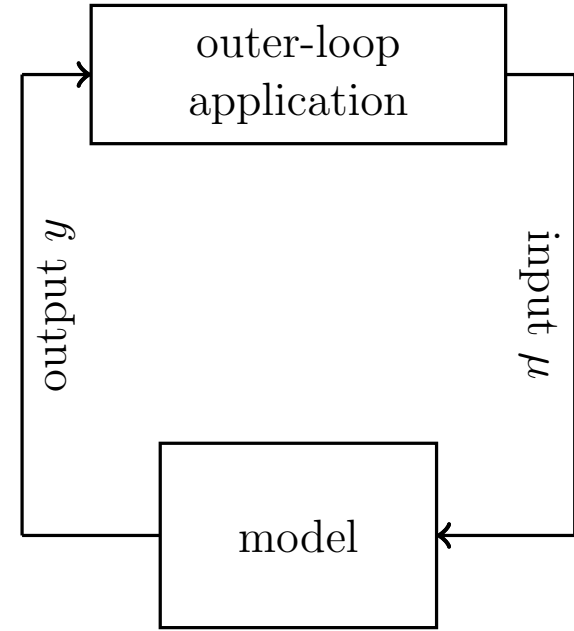
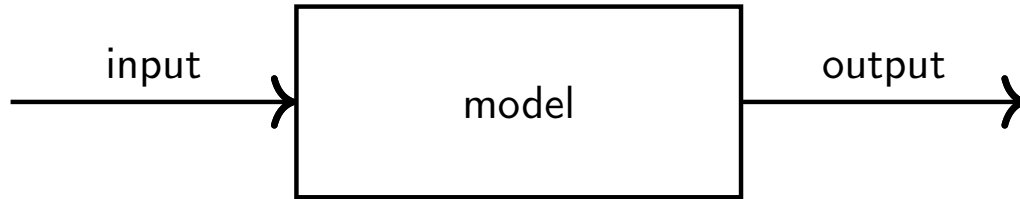
Topics that we will cover

- Proper orthogonal decomposition
- Reduced basis techniques
- Offline-online decompositions
- Error estimation
- Empirical interpolation
- Operator learning

Intro: Models and outer-loop applications

Model

- Model describes response of system to inputs, parameters
- Evaluating a model requires numerical simulations
- Many models described as differential equations

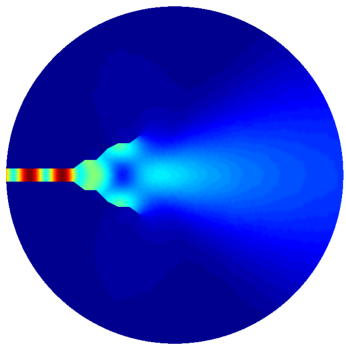


Outer loop applications

- Form outer loops around a model
- In each iteration an input μ is received and the corresponding model output y is computed
- An overall outer loop result is obtained at the termination of the outer loop

Challenge: Single model solve expensive; repeated solves in outer loop prohibitive

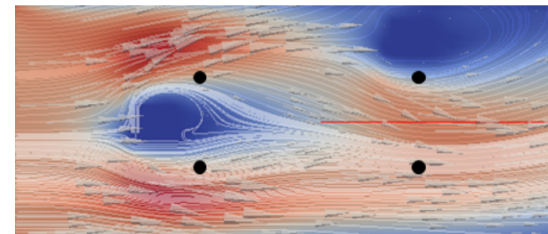
Intro: Outer-loop applications



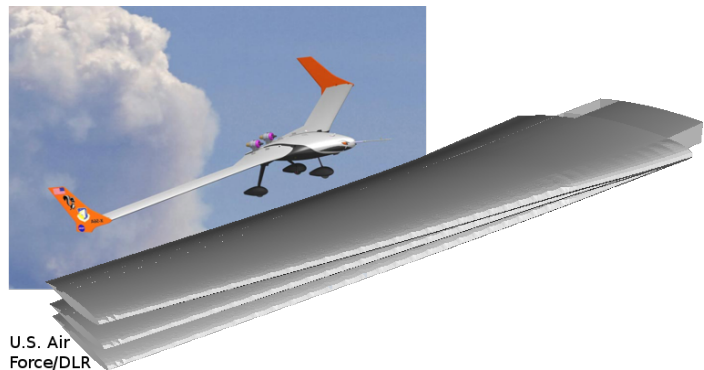
optimization



control



inference



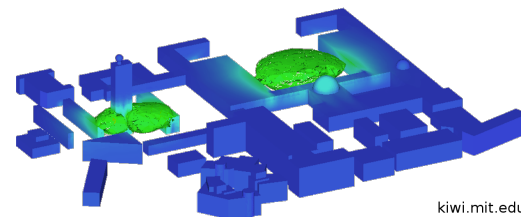
multi-discipline coupling



visualization



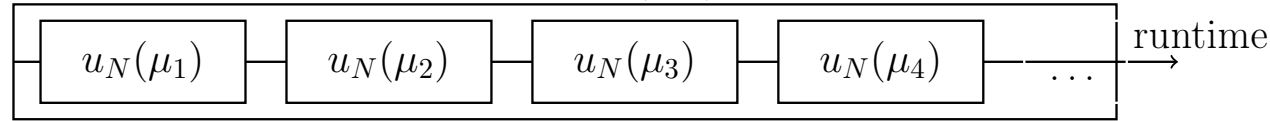
model calibration



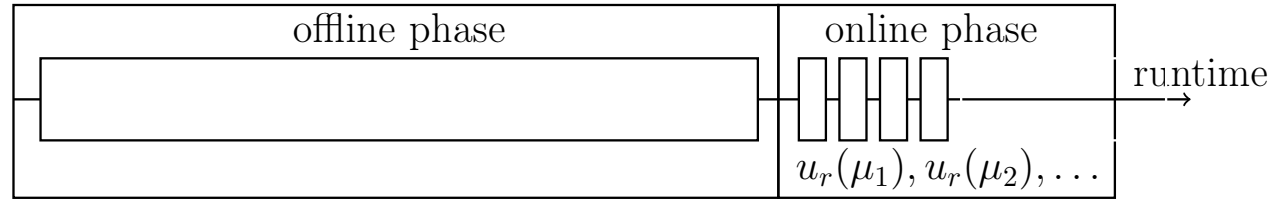
uncertainty quantification

Intro: Offline/online decomposition

Outer-loop application with high-fidelity (full) model:



Outer-loop application with surrogate model:



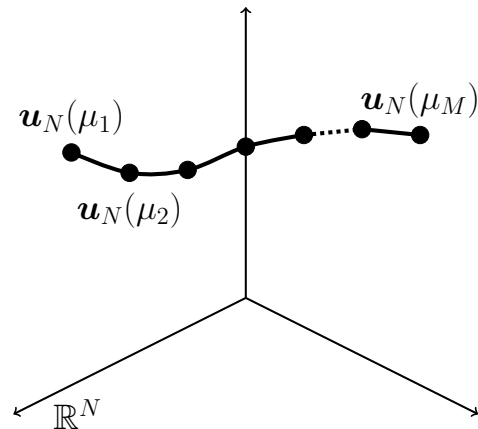
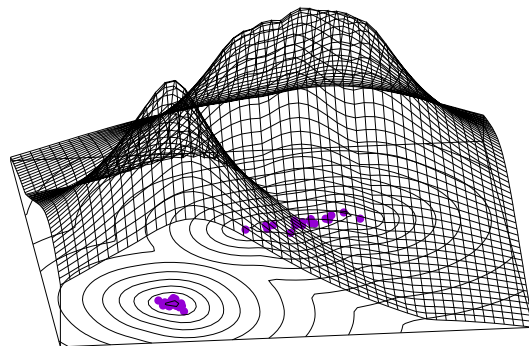
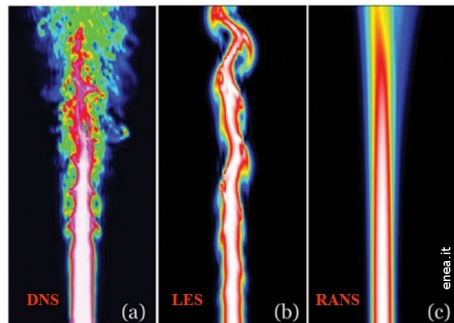
Offline (training) phase

- Generate snapshots (data) using the expensive, high-fidelity model
- Extract patterns from data and derive cheap surrogate model

Online (evaluation) phase

- Evaluate surrogate model instead of high-fidelity model (or both $\rightarrow$ multi-fidelity)
- Rapid prediction, control, optimization, uncertainty quantification

Intro: Three types of surrogate models



simplified surrogates

- Simplifying physics
- Coarser discretizations
- Linearized models
- Early stopping of iterative solvers

data-fit surrogates

- Fitting model to data of input-output map given by high-fidelity model
- Gaussian processes
- Neural networks
- Operator learning

reduced models

- Extract important dynamics of full-model states from *data*
- Approximate high-dimensional states in subspaces
- Restrict solving governing equations to subspaces

Plan for the rest of today

1. Low-dimensional structure
2. Reduced-space construction
3. Projection-based reduced models
4. Online efficiency
5. Challenges and course outlook

Full model

$$A(\mu) u_N(\mu) = F(\mu)$$

$$A(\mu) \in \mathbb{R}^{N \times N}$$

$$u_N(\mu) \in \mathbb{R}^N$$

$$F(\mu) \in \mathbb{R}^N$$

→ state $u_N(\mu)$

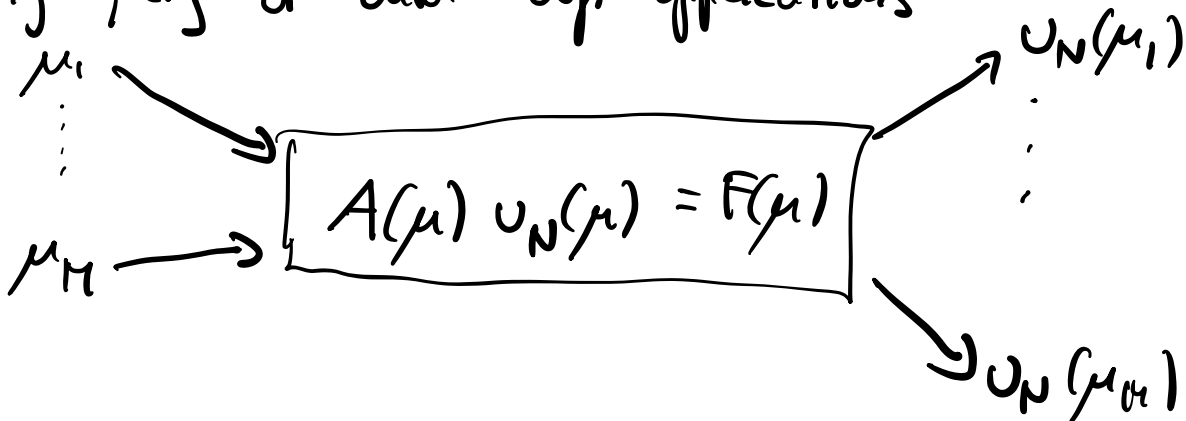
→ Dimension of state: $N \in \mathbb{N}$

→ Parameter: $\mu \in D \subseteq \mathbb{R}^p$

→ System matrix $A(\mu)$ - invertible for all $\mu \in D$

→ Right-hand side $F(\mu)$

Many-query or outer-loop applications



Costs of solving linear system scales with N .

Typically setting: $N \gg 1$, $M \gg 1$

→ one full-model solve already expensive

→ M many becomes prohibitive

But: we only vary parameter μ !

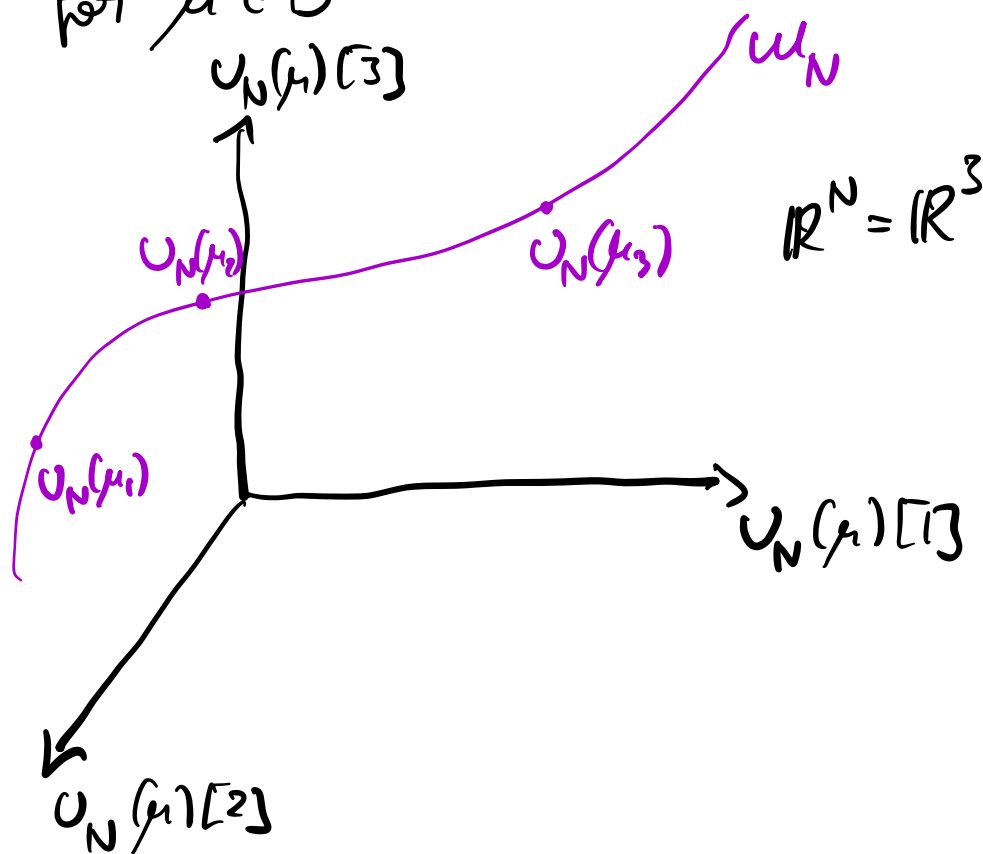
Solution manifold

$$\mathcal{U}_N = \{u_N(\mu) \mid \mu \in D\} \subset \mathbb{R}^N$$

(not necessarily a manifold - just a set)

→ contains all solutions of $A(\mu)u_N(\mu) = F(\mu)$

for $\mu \in D$



Ambient dimension N can be high, even though solutions $u_N(\mu)$ may "occupy" only "small" part of ambient space.

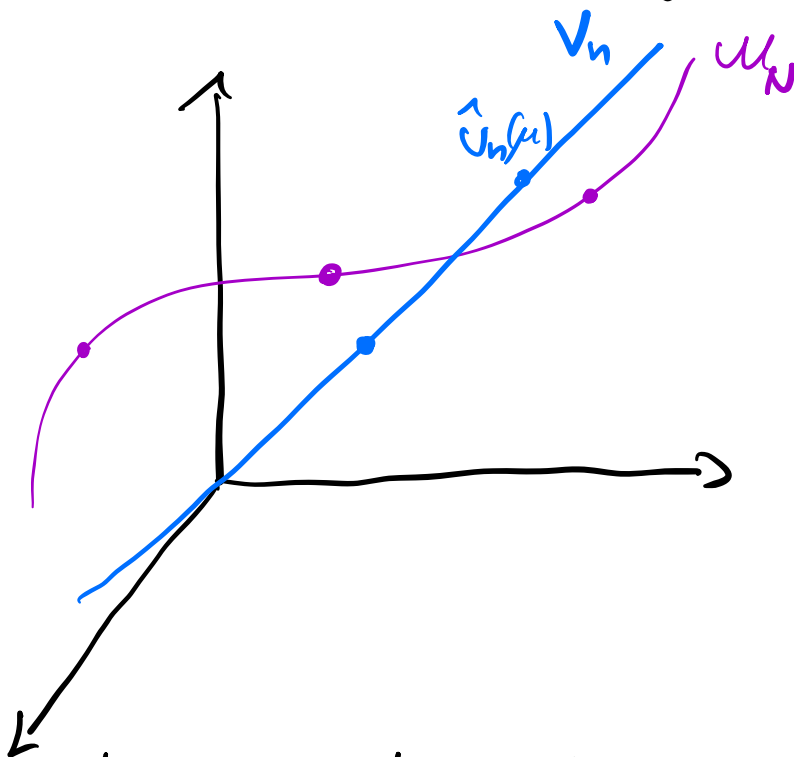
Model reduction: exploits this low-dim. structure!

Model reduction (classical linear model reduction)

↳ Seek subspace $V_n \subset \mathbb{R}^N$ of $\dim(V_n) = n$ with $n \ll N$ that "approximates well" the elements of the solution manifold $\mathcal{U}_N$.

↳ Then seek approximation $\hat{u}_n(\mu) \in V_n$ of solution of full model for given $\mu \in D$.

→ solution depends on $n \ll N$ degrees of freedom $\Rightarrow$ possibly quicker computation



↳ reduced space V_n is independent of μ

↳ tension btw low dimension n and low error
$$\|u_N(\mu) - \hat{u}_n(\mu)\|$$

for $\mu \in D$.

Parameter dimension vs. reduced space dimension

Consider $\mu \in [0,1] \subset \mathbb{R}^P$, $\underline{P=1}$

$$A(\mu) = \begin{bmatrix} 1+\mu & 0 \\ 0 & 1+2\mu \end{bmatrix}, \quad F = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$v(\mu) = A(\mu)^{-1}F = \begin{bmatrix} (1+\mu)^{-1} \\ (1+2\mu)^{-1} \end{bmatrix}$$

$$v(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v(1) = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \end{bmatrix}$$

are linearly independent

$\Rightarrow p=1$ need not imply exact representation in one-dim. subspace.

Instead consider the Kolmogorov n -width:

$$d_n(\underline{\mathcal{U}}_N) = \inf_{\substack{V_n \subset \mathbb{R}^N \\ \dim(V_n)=n}} \sup_{\mu \in D} \inf_{v \in V_n} \|v_n(\mu) - v\|$$

$\hookrightarrow$ smallest worst-case error achievable by any n -dim subspace.

$\hookrightarrow$ Property of solution manifold $\mathcal{U}_N$, not of any specific algorithm

Basic requirement for (classical, linear) model reduction:
 $d_n(\mathcal{U}_N)$ decays quickly with n
(=exp. decay)

Snapshots:

↳ A Kolmogorov-optimal subspace is typically not available $\leadsto$ approximate one.

↳ Consider a finite set that approximately represents $\mathcal{U}_N$

$$\{u_N(\mu_1), \dots, u_N(\mu_m)\} \subset \mathcal{U}_N$$

→ training set

→ provides information about $\mathcal{U}_N$

→ Quality of what follows can only be as good as training data

↳ then consider for example

$$\arg \min_{\substack{V_h \subset \mathbb{R}^N \\ \dim(V_h)=n}} \frac{1}{m} \sum_{i=1}^m \min_{v \in V_h} \|u_N(\mu_i) - v\|_2^2 \quad (*)$$

→ find best n -dim subspace that approximates training data in terms of (*)

Collect snapshots into the snapshot matrix

$$U = \begin{bmatrix} | & & | \\ U_N(\mu_1) & \dots & U_N(\mu_n) \\ | & & | \end{bmatrix} \in \mathbb{R}^{N \times n}$$

Compute SVD of U

$$\text{rank}(U) = r$$

$$U = \underline{V} \Sigma W^T, \quad \sigma_1 \geq \dots \geq \sigma_r > 0$$

$$V = \begin{bmatrix} | & \dots & | \\ v_1 & \dots & v_r \\ | & & | \end{bmatrix} \in \mathbb{R}^{N \times r}$$

Reduced space of dimension $n \in \mathbb{R}$ is

$$\underline{V}_n = [v_1, \dots, v_n] \in \mathbb{R}^{N \times n} \quad \begin{array}{l} \underline{V}_n \dots \text{basis matrix} \\ v_n \dots \text{space} \end{array}$$

Corresponding projection error

$$\frac{1}{n} \sum_{j=1}^n \| U_N(\mu_j) - \underline{V}_n \underline{V}_n^T U_N(\mu_j) \|_2^2 = \frac{1}{n} \sum_{j=n+1}^r \sigma_j^2$$

↳ Often called proper orthogonal decomposition (POD)

↳ „Best“ subspace only for training data

↳ Decay of singular values indicate
low-rank structure

→ snapshots are key

From basis to reduced model

↳ For new $\mu \in D$ ($\neq$ not in training data), seek an approximation of full solution in V_h

↳ All elements of V_h can be represented as

$$\textcircled{*} \quad \hat{u}_h = \underline{V}_h \underline{u}_h \in V_h, \quad \underline{u}_h \in \mathbb{R}^n$$

$\underline{u}_h$... coefficient vector

↳ Insert $\textcircled{*}$ into full model to get residual

$$r_N(\underline{u}_h; \mu) = F(\mu) - A(\mu) \underline{V}_h \underline{u}_h(\mu)$$

→ n unknowns in N equations

Galerkin condition: find $\underline{u}_h$ so that residual is orthogonal to V_h

$$r_N(\underline{u}_h; \mu) \perp V_h$$

$$\underline{v}_j^T r_N(\underline{u}_h; \mu) = 0, \quad j=1, \dots, n$$

$$\underline{V}_h^T r_N(\underline{u}_h; \mu) = 0$$

Write out residual

$$\underline{V}_n^T [F(\mu) - A(\mu) \underline{V}_n \underline{u}_n(\mu)] = 0$$

$$\underbrace{\underline{V}_n^T A(\mu) \underline{V}_n}_{= A_n(\mu) \in \mathbb{R}^{n \times n}} \underline{u}_n(\mu) = \underbrace{\underline{V}_n^T F(\mu)}_{= F_n(\mu) \in \mathbb{R}^n}$$

$$A_n(\mu) \underline{u}_n(\mu) = F_n(\mu)$$

Interpretation of Galerkin as best approximation

↳ Assume $A(\mu)$ is sym. pos. def (SPD)

↳ Define error

$$e(\mu) = u_N(\mu) - V_n u_n(\mu)$$

↳ Residual

$$\begin{aligned} r_N(u_n(\mu); \mu) &= F(\mu) - A(\mu) V_n u_n(\mu) \\ &= A(\mu) u_N(\mu) - A(\mu) V_n u_n(\mu) \\ &= A(\mu) e(\mu) \end{aligned}$$

↳ Galerkin condition

$$0 = V_n^T r(u_n(\mu); \mu) = V_n^T A(\mu) e(\mu)$$

↳ Because $A(\mu)$ is spd, we have inner product: $\langle x, y \rangle_{A(\mu)} = x^T A(\mu) y$

→ Galerkin solution $u_h(\mu)$ gives

$$e(\mu) \perp_{A(\mu)} V_h$$